

6

Order Effects Within a Question

Presenting Categorical Response Alternatives

In this chapter we turn to the impact of the order in which different response alternatives are presented within a single question. As is the case for question order effects, survey researchers have long been aware that the order in which response alternatives are presented to respondents can profoundly affect the obtained results (see Blankenship, 1943; Cantril, 1944; Payne, 1951, for early discussions). Empirically, *primacy effects* (that is, higher endorsements of the same items if presented early in the list) and *recency effects* (higher endorsements of the same items if presented late in the list) have been obtained in numerous studies. The emergence of response order effects is apparently not restricted to one response format and has been observed on verbal rating scales (for example, Carp, 1974) as well as on forced-choice questions that present a small number (McClendon, 1986) or a large number (Becker, 1954) of discrete response alternatives. Moreover, response order effects have been obtained on opinion questions as well as on factual reports (Schwarz, Hippler, & Noelle-Neumann, 1992, 1994) and they are not limited to survey results. The order in which alternatives are presented has also been shown to influence decisions with potentially important consequences. For example, a political candidate may receive more votes when his or her name is presented early rather than late on the ballot sheet, reflecting a primacy effect (Brook & Upton, 1974; Mueller, 1970).

Despite the considerable empirical evidence for the emergence of response order effects, we know relatively little about the conditions that determine their emergence and direction, and the area is characterized by a large number of apparently inconsistent findings. These inconsistencies presumably indicate that different processes may contribute to the emergence of response order effects. Moreover, these processes are not mutually exclusive, resulting in a rich set of possible interaction effects. Unfortunately, the presumably systematic character of these interaction effects can hardly be detected in the available studies because they are usually restricted to a manipulation of only one or two factors.

Complicating things further, the most frequently used research designs show a surprising lack of sophistication, rendering it impossible to isolate even the impact of the key variable, namely, presentation order per se. Let us suppose that a researcher presents three response alternatives. To explore a possible order effect, he reverses the presentation order for half of the sample, resulting in a Form A (ABC) and a Form B (CBA). In this typical design, presentation order is confounded with the content of preceding response alternatives. As a result, it is impossible to decide if differences in the endorsement of A are due to A's serial position or to the possible influence of thinking about C or B before being exposed to A. To isolate the impact of presentation order alone, one would need to employ a Latin square design, in which each response alternative occurs once in each ordinal position. In this example, such a design would result in six (rather than two) forms: ABC, ACB, BAC, BCA, CAB, and CBA. Whereas the use of Latin square designs is a standard procedure in experimental psychology (see Lindquist, 1953, pp. 273–281, for an early textbook introduction), adequate designs have rarely been used in survey research on response order effects because they are complex to administer (see Becker, 1954, for a noteworthy exception).

Because of these shortcomings in the available data, it is currently impossible to draw strong conclusions about the processes that underlie response order effects in survey measurement. Therefore,

our goal for the present chapter is to review different process assumptions in the light of current psychological theorizing, highlighting plausible hypotheses that need to be tested in future research. Not surprisingly, our own favorite account of response order effects relies on the conceptual framework that we developed in the two preceding chapters and focuses on the impact of response alternatives on the cognitive accessibility of relevant information and its use in the construction of mental representations of the target. The reader should keep in mind, however, that much of our discussion is speculative. Although we pay close attention to the compatibility of our theorizing with the available data, the nature of these data does not allow unequivocal conclusions.

What Mediates Response Order Effects?

Different explanations for the emergence of response order effects have been offered over the decades. Surprisingly, none of the early hypotheses has suffered in popularity, reflecting the limited progress made in this area. One explanation assumes that respondents "like to reach for the first thing that catches their eye" (Payne, 1951, p. 83), an assumption that is also at the heart of Krosnick's (1991) discussion of satisficing in survey measurement. Hence, any item "that catches the eye" should be more likely to be endorsed if presented early rather than late. Accordingly, this notion predicts primacy effects and cannot account for the emergence of recency effects, rendering it unsuitable as a general explanation. A second account attributes response order effects to respondents' memory limitations (for example, Blankenship, 1943). As we show in detail in the following section, this explanation predicts recency effects under conditions that are typical for survey interviews and cannot account for primacy effects unless one makes questionable additional assumptions.

A more recent line of theorizing suggests as a third possibility that the presentation order influences respondents' opportunities to think about a given response alternative, thus influencing the judg-

ment stage of the question answering sequence (Krosnick, 1992; Krosnick & Alwin, 1987; Schwarz, Hippler, & Noelle-Neumann, 1992, 1994). This cognitive elaboration account predicts a number of complex interactions of presentation order and presentation mode (that is, whether the response alternatives are presented in writing or are read to respondents) that are consistent with much of the available data.

Finally, some response order effects do not result from serial position *per se* but reflect an impact of the content of earlier items, as we discussed in the preceding chapters. We address each of these accounts in more detail in the following section.

Other hypotheses offered in the literature do not provide causal accounts of the emergence of response order effects but address limiting conditions. Many researchers assumed that response order effects are more likely to emerge when the questions are complex (Payne, 1949). This hypothesis is compatible with the memory limitation as well as the cognitive elaboration accounts. In addition, it has frequently been assumed that response order effects are not obtained when respondents' opinions are "crystallized" (Rugg & Cantril, 1942). This hypothesis is compatible with the cognitive elaboration account to be presented in this chapter, if we assume that "crystallization" implies that a considerable amount of relevant information is chronically accessible to respondents, limiting the impact of temporarily accessible information brought to mind by response alternatives. Similarly, the order of response alternatives should have little impact if respondents can recall a previously formed judgment, provided that this judgment matches one of the response alternatives offered. These considerations parallel our discussion of the emergence of question order effects in Chapter Five. In contrast, sheer familiarity with the subject matter is insufficient to eliminate order effects, as Becker (1954) observed in a study that assessed evaluations of well-known radio programs. In line with our previous discussion of the crystallization issue (see Chapter Five), Becker's findings indicate that having well-developed attitudes based on extensive experience with the subject matter is not

sufficient to eliminate judgmental context effects unless a previous judgment is highly accessible.

We now turn to a detailed discussion of *memory limitations*, *cognitive elaboration*, and *contrast effects*. We focus on questions that present a number of discrete categorical response alternatives. Throughout, our emphasis is on the likely emergence of complex but systematic interaction effects. In the final section of the chapter, we address a special case, the emergence of order effects on verbal rating scales.

Memory Limitations

Response order effects have frequently been attributed to respondents' memory limitations. If respondents have forgotten a response alternative by the time they are supposed to report their answer, that alternative will not be endorsed. According to this theory, respondents' selection is restricted by their memory performance and a given response alternative is more likely to be endorsed if it is more likely to be recalled. Not surprisingly, psychological research into the learning of lists of verbal expressions indicates that recall is indeed influenced by the length of the list and by an item's serial position (see Baddeley, 1990; Smyth, Morris, Levy, & Ellis, 1987, for reviews). However, the direction of this influence is highly dependent on the time delay between learning and recall, a contingency that has frequently been overlooked in attempts to construct memory-based accounts of presentation order effects.

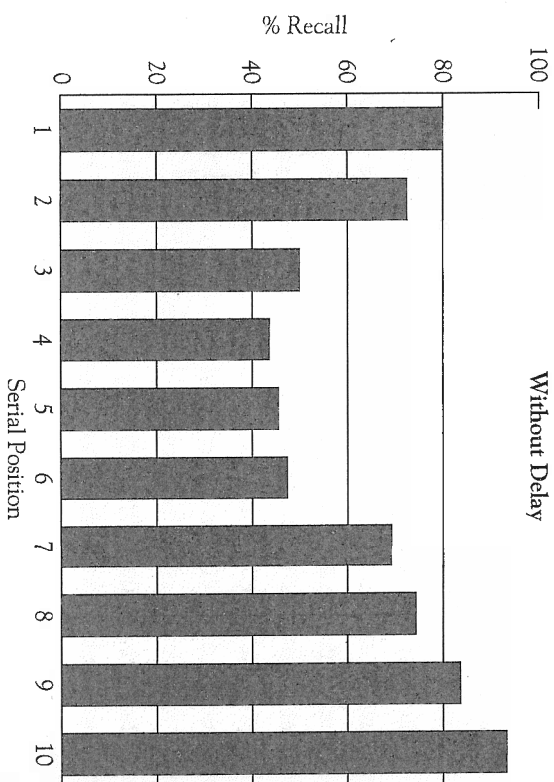
The left-hand panel of Figure 6.1 shows a typical serial position curve under conditions of immediate recall, that is, conditions under which recall follows learning without delay. Under these no-delay conditions, material presented at the end of the list is most likely to be recalled because the last few items are still accessible in working memory. However, the material presented at the beginning of the list is more likely to be recalled than material presented in the middle of the list even though more time has elapsed since it

was learned. This reflects the fact that the first few items suffer "less competition for time and space" (Smyth et al., 1987, p. 123) in working memory than do later items. They are therefore more extensively processed and more likely to be committed to long-term memory. Recall is poorest for items presented in the middle of the list. These items are less likely to be committed to long-term memory, putting them at a disadvantage relative to the earliest items, and they are also no longer present in working memory, putting them at a disadvantage relative to the last few items. As a result, experimental research in the list learning tradition has typically obtained recency effects under conditions of immediate recall, indicating that recall is highest for the last few items.

Next, let us consider a typical serial position curve under conditions of delayed recall. An example is shown in the right-hand panel of Figure 6.1. Here, items presented at the beginning of the list are most likely to be recalled, indicating that these items are committed to long-term memory from where they can be recalled even after time has elapsed. The recall advantage of the last few items, however, can no longer be observed. Because of the delay, these items have been cleared from working memory, in particular if subjects had to work on a distractor task during the delay. Therefore, recall performance on the last items approaches the low level of the middle items, reflecting their disadvantaged status in long-term memory. Accordingly, list learning experiments have typically produced primacy effects under conditions of delayed recall.

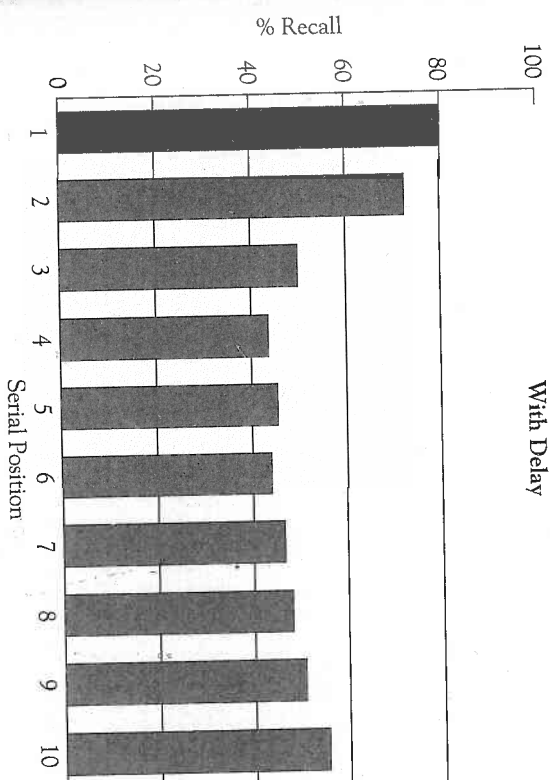
What are the implications of this well-established interaction effect between serial position and time of recall for response order effects in survey measurement? These considerations suggest that we should typically observe the emergence of recency effects in long lists of response alternatives. Because survey respondents report their answer immediately after exposure to the response alternatives, there is no delay between "learning" and "recall." Accordingly, response alternatives presented at the end of the list should be easily accessible in working memory. Because under no-

FIGURE 6.1 Recall as a function of serial position and delay.



delay conditions recall of late items from working memory is better than recall of early items from long-term memory, the result should be pronounced recency effects, much as is shown in the left-hand panel of Figure 6.1. Primacy effects should only be obtained if a delay is introduced between exposure to the response alternatives and respondents' reports, which is not usually the case in survey interviews. Nevertheless, primacy rather than recency effects have usually been reported in survey experiments with long lists of response alternatives (Payne, 1951; Becker, 1954; Krosnick & Alwin, 1987; Mueller, 1970; Ring, 1975). We conclude from this inconsistency between the two literatures that, despite popular assumptions, sheer memory limitations are not the primary source of response order effects in survey measurement.

In fact, long lists of response alternatives are usually presented on show cards that remain visible until respondents report their answer, thus putting little burden on their memory to begin with. With telephone interviews the problem may be more acute because

FIGURE 6.1 Recall as a function of serial position and delay (*continued*).

show cards cannot be used. However, response order effects have also been observed on questions that present only two or three response alternatives (Payne, 1951; McClendon, 1986; Schuman & Presser, 1981; Schwarz, Hippler, & Noelle-Neumann, 1992), and we will present some examples later in this chapter. Such a limited number of response alternatives should be easily accessible in working memory under the no-delay condition of survey interviews, thus rendering the emergence of order effects unlikely unless the alternatives are overly complex. Accordingly, it seems that we need to consider other processes if we want to explain response order effects in survey measurement.

In summary, memory limitations are most likely to play a role in the emergence of response order effects when numerous or complex response alternatives are presented without the help of show cards, thus taxing respondents' memory. Under this condition, psychological theorizing about memory limitations predicts the emergence of recency effects. Primacy effects would only be pre-

dicted under delayed recall conditions, which are atypical for survey interviews.

Cognitive Elaboration

If memory limitations do not provide a satisfactory explanation for most response order effects in survey measurement, what other processes should be considered? Following previous theorizing by Krosnick and Alwin (1987) and drawing on cognitive research in the area of persuasive communication (for example, Eagly & Chaiken, 1993; Petty & Cacioppo, 1986a, 1986b), one might expect that the order and mode in which response alternatives are presented affect respondents' opportunity to elaborate on their content. This effect, in turn, influences the thoughts they generate in response to opinion questions and the quality of retrieval cues they generate in response to behavioral questions. Such a model has been proposed by Schwarz, Hippler, and Noelle-Neumann (1992; 1994), and our current discussion draws on their theorizing. According to their model, the impact of an item's serial position is crucially dependent on whether it is presented in a visual or an auditory mode. Furthermore, the mediating process is the respondent's opportunity to think about the item's content. We first consider the implications of the cognitive elaboration approach for opinion questions and then turn to its implications for autobiographical reports.

Response Alternatives as Persuasive Arguments

Let us suppose that a respondent asked an opinion question is provided with several response alternatives that reflect different views. We may think of each response alternative as a short persuasive argument. Hence, the processes that are known to determine the impact of persuasive communications may also determine respondents' reactions to the items presented to them. As a large body of

research has demonstrated, the impact of a persuasive argument is determined to a large degree by the recipient's cognitive responses, that is, by the thoughts that the recipient generates in response to the presented argument (see Eagly & Chaiken, 1993; Petty & Cacioppo, 1986a, 1986b, for reviews). The more positive, agreeing thoughts the argument elicits, the more the recipient is influenced by it, resulting in a positive attitude change. Conversely, if the argument elicits negative, disagreeing thoughts, the recipient is likely to exhibit a "boomerang" attitude change, moving away from the argument's implications.

These assumptions are compatible with Schwarz and Bless's (1992a) inclusion/exclusion model, which we reviewed in Chapter Five. Specifically, the agreeing or disagreeing thoughts that the response alternatives bring to mind are likely to become part of the mental representation that respondents form of the proposition offered in the response alternative. In general, the more agreeing thoughts a response alternative elicits, the more likely the respondents are to endorse it. Conversely, the more disagreeing thoughts a response alternative elicits, the less likely the respondents are to endorse it. The number of thoughts elicited by a given response alternative, however, is not only a function of the content of the response alternative. Thoughts are also influenced by the recipients' ability and motivation to process the content and by the opportunity that the situation provides them to do so. As research on persuasive communications has demonstrated (see Eagly & Chaiken, 1993; Petty & Cacioppo, 1986a, 1986b for reviews), a plausible argument is more likely to elicit agreement the more the respondent thinks about its implications whereas an implausible argument is more likely to elicit disagreement under these conditions. Thus, conditions that interfere with recipients' cognitive elaboration of the implications of a presented argument, such as distractions while they listen to the message, have been shown to eliminate the advantage of plausible over implausible arguments (Bless, Bohner, Schwarz, & Strack, 1990; Harkins & Petty, 1981; see Petty & Brock, 1981 for a review).

How do these considerations relate to response order effects? As Krosnick and Alwin (1987, see also Krosnick, 1992) have argued in a very persuasive paper that strongly influenced Schwarz et al.'s (1992) theorizing, the degree of cognitive elaboration that a given response alternative receives is likely to be influenced by the order and the mode in which the response alternatives are presented. Let us suppose that a long list of response alternatives is presented to respondents in a visual format, either on a show card as part of a face-to-face interview or in a self-administered questionnaire. Under these conditions, "Items presented early in a list are likely to be subjected to deeper cognitive processing," as Krosnick and Alwin (1987, p. 213) noted. "By the time a respondent considers the later alternatives, his or her mind is likely to be cluttered with thoughts about previous alternatives that inhibit extensive consideration of later ones." As a result, visual presentation formats facilitate the cognitive elaboration of items presented early in the list and inhibit the elaboration of those presented late in the list.

However, if the items are not presented visually but instead are read to respondents by the interviewer, either under face-to-face or telephone interview conditions, respondents have little opportunity to elaborate on the items presented early in the list. The time available for processing is restricted by the speed with which the interviewer moves on to the next item. "Under these circumstances, respondents are able to devote most processing time to the final item(s) read, since interviewers usually pause most after reading them" (Krosnick & Alwin, 1987, p. 203). In addition, respondents may find it difficult to keep all the response alternatives in mind without visual cues, as discussed earlier. Accordingly, auditory presentation formats facilitate the cognitive elaboration of items presented late in the list and inhibit the elaboration of items presented early in the list.

However, as we have seen, the impact of the opportunity to elaborate on the implications of a response alternative depends on the alternative's plausibility. In operational terms, a given item is "plausible" to a respondent if it elicits agreeing thoughts when the respondent thinks about it and "implausible" if it elicits disagree-

ing thoughts, paralleling the conceptualization of agreeing or disagreeing "cognitive responses" in persuasion research (see Eagly & Chaiken, 1993; Petty & Cacioppo, 1986a, 1986b). In combination with the impact of the visual and auditory presentation formats on a respondent's opportunity to think about items presented early or late in a list, these considerations predict a complex pattern of interaction effects.

If a visual presentation format is used, respondents are more likely to elaborate on the items presented early in the list, for the reasons already discussed. Accordingly, a given plausible item should elicit more agreeing thoughts and be more likely to be endorsed when it is presented early rather than late in the list, resulting in a primacy effect. Conversely, a given implausible item should elicit more disagreeing thoughts and be less likely to be endorsed when presented early rather than late in the list, resulting in a recency effect.

The opposite predictions hold for auditory presentation formats. As noted, in an auditory format items presented late in the list are likely to receive more elaborate processing than items presented early. Hence, a plausible item is likely to elicit more agreeing thoughts and is more likely to be endorsed if presented late in the list, resulting in a recency effect. Conversely, an implausible item is likely to elicit more disagreeing thoughts and is less likely to be endorsed if presented late in the list, resulting in a primacy effect.

Altogether, these considerations predict a triple interaction of serial position, presentation mode, and item plausibility. Assuming that an item is plausible, thus eliciting agreeing thoughts, primacy effects should emerge under a visual presentation mode and recency effects under an auditory presentation mode. On the other hand, if the item is implausible the opposite predictions hold. In the latter case, recency effects should emerge under a visual presentation mode and primacy effects under an auditory presentation mode.

As complex as these predictions are, an additional factor needs to be considered. As Krosnick (1992) noted, respondents' thoughts about a response alternative may exhibit a confirmatory bias. For example, a respondent who is asked which of a list of problems is

more important may be more likely to search for reasons that make a given problem important than unimportant. In fact, the task of identifying important problems offers a directed search hypothesis, and numerous studies have shown that such directed searches favor the retrieval of confirmatory material (see Wyer & Srull, 1989, for a review). This confirmation bias should facilitate the generation of agreeing thoughts in response to plausible items and inhibit the generation of disagreeing thoughts in response to implausible items. As a result, the predicted response order effects should be more pronounced in the former than in the latter case. In other words, we should be more likely to find support for our predictions concerning plausible rather than implausible items. In addition, the impact of cognitive elaboration differences is most likely to be observed under conditions that do not tax respondents' memory, thus limiting the possible impact of the recall processes discussed, which may dilute elaboration effects.

In the next section we review the limited experimental evidence available.

Opinion Questions: Experimental Evidence

Unfortunately, well-controlled experimental studies that test the theoretically crucial triple interaction of serial position, presentation mode, and item plausibility are not yet available. With few exceptions, the available evidence is limited to studies that were not originally intended to test cognitive elaboration assumptions. Moreover, each of the available studies bears only on some of the variables, and none of the studies contains direct manipulations of item plausibility or measures of the assumed underlying cognitive processes. Given these limitations, however, it is encouraging to note that archival data, reviewed by Schwarz, Hippler, and Noelle-Neumann (1992, 1994), are largely consistent with predictions.

Presentation Order and Mode. Most studies that obtained primacy effects in long lists of response alternatives were presented in

a visual format (for example, Becker, 1954; Krosnick & Alwin, 1987; Schwarz, Hippler, and Noelle-Neumann, 1992). This is consistent with the model's prediction of primacy effects under visual presentation conditions (assuming plausibility of the items). However, data bearing on the auditory presentation of long lists are missing, because long lists are usually presented visually to avoid memory problems. Thus, the available archival data allow no evaluation of the predicted recency effects under auditory mode conditions.

More important, analyses of split-ballot experiments that used questions with only two or three response alternatives found support for the predicted serial position times mode interaction. Specifically, secondary analyses of split-ballot experiments originally conducted by Elisabeth Noelle-Neumann of the Allensbach Institute in Germany revealed a preponderance of primacy effects if the response alternatives were presented on show cards but a preponderance of recency effects if they were read to respondents (see Schwarz, Hippler, and Noelle-Neumann, 1992, for details). These effects were obtained with complex as well as relatively simple response alternatives. Table 6.1 shows some examples. Other researchers who explored order effects with two or three response alternatives also obtained recency effects (Krosnick, 1992; McClendon, 1986; Schuman & Presser, 1981) when using an auditory presentation mode. It should be noted that the presentation of only two or three simple alternatives is unlikely to tax respondents' memory, further supporting the notion that these effects are likely to reflect differences in cognitive elaboration rather than sheer memory limitations.

Judgment Formation. Another implication of the cognitive elaboration assumption is that response order effects should be obtained only when respondents need to form a judgment on the spot, as our discussion in the preceding chapters would suggest. In such cases, their judgment should be influenced by their opportunity to elaborate on the response alternatives. If respondents can recall a chronically accessible judgment from memory, however, the impact of order and mode should be largely reduced. Similarly, differences

TABLE 6.1 Response order effects in dichotomous questions.

Presentation mode	Response order	
	Presented first	Presented second
<i>Visual</i>		
Example 1:		
Responsibility for welfare		
State	64%	50%
Private charity	34%	28%
<i>Auditory</i>		
Example 2:		
Form of government		
Authoritarian	15%	26%
Democratic	58%	67%
Example 3:		
Preferred novel		
Serious	31%	35%
Humorous	48%	53%

Question wordings and data sources

Example 1: Responsibility for welfare

Presentation order of statements on show card is manipulated.

"Here are two men talking with one another. Would you please read what the two of them say?" (Interviewer gives respondent time to read show card.) "Which of the two would you rather agree with? With the upper one or with the lower one?"

Show card presents two individuals with the following statements in speech bubbles:

"I feel it is the responsibility of the state to make sure that there is no social misery. Retirement payments, social welfare, and sick leave compensation must be sufficient for one's living. That is the only adequate way to eliminate social misery."

"I feel that the state cannot ease all social misery. There are many individual cases where state payments are not all that is needed. One often needs to consider the individual circumstances. Therefore, each individual should help through donations and personal commitment to ease social misery."

Source: HB 1064, Institut für Demoskopie, Allensbach.

TABLE 6.1 Response order effects in dichotomous questions (continued).

Example 2: Form of government

Presentation order of statements manipulated; no show card used.

"Two men talk about how a country should be governed.

One says: I prefer that the people put its best politician at the top and delegates all power to him. Then he and selected experts can make clear and fast decisions. There isn't much talking and things are really going to happen.

The other says: I prefer that several people have some influence in the state. That results in some talking before anything gets done but it makes sure that power cannot be abused that easily.

Which of these two opinions is closer to your own—the first one or the second one?"

Source: HB 1044, July 1960, Institut für Demoskopie, Allensbach.

Example 3: Preferred novel

Presentation order of "serious"/"humorous" manipulated.

"If you had the choice to read either a serious or a humorous novel, what would you rather read these days: the serious or the humorous novel?"

Source: HB 445, March 1957, Institut für Demoskopie, Allensbach.

in the elaboration of the response alternatives should be less influential if respondents elaborate on the same content domain while answering preceding questions, increasing the temporary cognitive accessibility of relevant thoughts. In sum, response order effects should be less pronounced when respondents have already formed a judgment that they can recall from memory or when preceding questions bear on related issues. Experimental findings support these hypotheses.

Schwarz, Hippler, and Noelle-Neumann (1992) conducted a

modified replication of Payne's (1951) "oil supply" experiment (see also Schuman & Presser, 1981). In their study, ninety-one adult citizens of Mannheim, Germany, were read the question, "Some people say that we will still have plenty of oil twenty-five years from now. Others say that at the rate we are using our oil, it will all be used up in about fifteen years. Which of these ideas would you guess is most nearly right?" The order in which the two opinions were presented was reversed for half of the sample. (See Table 6.2.)

As shown in the first part of the table, a pronounced recency effect of 33 percentage points was obtained for both response alternatives when presented in an auditory format, replicating previous findings. For half of the sample, however, the question was preceded by two other questions that tapped the same content domain, which should have triggered cognitive elaborations bearing on the oil supply issue. These questions concerned the respondents' attitudes toward restrictions in the consumption of oil and the development of alternative sources of energy. As expected, introducing these context questions not only affected respondents' overall attitudes but completely eliminated the response order effect, as shown in the second part of Table 6.2.

This finding supports the general hypothesis that response order effects are in part a function of the cognitive elaboration of the response alternatives: if respondents are induced by preceding questions to elaborate on the issue before being exposed to the response alternatives, response order effects may be reduced or even eliminated. At the same time, the impact of the preceding questions on respondents' reports indicates that having respondents elaborate on the content domain may elicit pronounced shifts in reported attitudes, as discussed in Chapter Five. Although this finding bears on the theoretical rationale offered here it is not intended to imply that researchers should replace response order effects by question order effects.

Some archival data are also consistent with the assumption that response order effects occur at the judgment stage and are reduced or eliminated if respondents can recall a previously formed judgment.

TABLE 6.2 The elimination of response order effects.

	Response order	
	Presented first	Presented second
Context		
Without preceding questions		
25 years	36%	69%
15 years	31%	64%
With preceding questions		
25 years	73%	71%
15 years	29%	27%

Note: The percentage of respondents who endorsed a given response alternative is reported. N = 91. Response alternatives were presented in an auditory format.

ment from memory. Table 6.3 shows one of the examples reported by Hippler, Schwarz, and Noelle-Neumann (1990), again taken from the archives of the Allensbach Institute.

In a split-ballot experiment conducted in February 1958, German respondents were asked to evaluate Chancellor Adenauer's performance on the issue of German reunification. The question was presented in a forced-choice form, offering two different opinions: (1) Adenauer had not done much on the issue, and (2) Adenauer could not do much because the Russians were not cooperative. As shown in the first part of Table 6.3, a recency effect of about 11 percentage points emerged on this dichotomous question, presented in an auditory format.

What should we expect if we break these data down by party preference? In the late 1950s, Adenauer was a highly respected political figure for conservative voters. Conservatives may not have needed to form a judgment when asked about Adenauer's performance on any issue, because the specific evaluation could easily be derived from their general evaluation of him, as we have discussed in preceding chapters. But respondents with a different party preference faced a more difficult task. As survey data from that time

indicate, most Germans who held a different party preference still greatly respected Adenauer but disagreed with him on some issues. Accordingly, these respondents presumably needed to consider the specific issue in determining their answer. Their responses were more likely to have depended on the specific thoughts elicited by the response alternatives. Indeed, the data shown in the second part of Table 6.3 are consistent with this assumption. Whereas conservatives' responses were not significantly affected by response order, those provided by respondents with a different party preference did show a recency effect of 13.5 percentage points.

Additional analyses showed that the emergence of these effects was not moderated by education. An apparent impact of frequency of church attendance was also eliminated when party preference was controlled for. Thus, attitudinal variables proved more important than sociodemographic variables, as we have previously noted in the context of question order effects. From an applied point of view, these findings indicate that the order in which the response alternatives are presented may have a differential impact on subsamples, adding to the complexities in this area.

TABLE 6.3 Response order effects as a function of necessary elaboration.

	Position	
	First	Second
<i>All respondents</i>		
Adenauer has not done much	32.7%	44.3%
Russians are uncooperative	55.7%	67.3%
<i>Party preference: CDU</i>		
Adenauer has not done much	12.8%	14.8%
Russians are uncooperative	85.2%	87.2%
<i>Party preference: Other</i>		
Adenauer has not done much	47.7%	61.2%
Russians are uncooperative	38.8%	52.3%

Source: Data based on a quota sample of German adults (IFD 1018, February 1958); DK excluded.

Note: Respondents were randomly assigned to one of the two presentation order conditions and response alternatives were presented in an auditory format.

Respondent Variables. Although no effect of level of education was observed in the preceding example, other researchers report that response order effects increase with decreasing education (Cochrane & Rokach, 1970; Krosnick & Alwin, 1987; McClelland, 1986) and decreasing cognitive sophistication, as measured by grade point average in a college sample (Krosnick, 1992). On theoretical grounds, however, we would not expect education per se to be a reliable moderating variable. Although respondents with high cognitive sophistication should find it easier to consider a variety of different arguments, this effect may be overridden by respondents' familiarity with the specific issue under study (which may determine the accessibility of relevant information) or by motivational variables. Depending on how these factors covary with education, the impact of education alone is likely to follow the familiar rule of "sometimes you see it, sometimes you don't."

Future Research. Most urgently needed in this domain are experiments that provide orthogonal manipulations of the three key variables (presentation order, presentation mode, and item plausibility) within the same study, using identical items in all order and mode conditions. Because the analysis of response order effects is necessarily based on frequency data and because the size of the largest of these effects rarely exceeds 10 percentage points, complex experiments with adequate power would require very large sample sizes; this explains the paucity of relevant studies. Although secondary analyses of archival data, as we have reviewed them in the preceding sections, support the general plausibility of the cognitive elaboration account, they cannot provide adequate tests of the underlying processes.

A Complication: Judgmental Contrast Effects

Response order effects are a function not only of respondents' memory limitations and the cognitive elaboration of the response alternatives provided but also of other judgmental processes

elicited by the items. To introduce these processes, we first have to highlight an important distinction between two types of response alternatives.

Dimensional and Nondimensional Response Alternatives

One type of response alternative consists simply of a list of heterogeneous possibilities. For example, respondents may be asked, "What did you do last Saturday?" and offered alternatives such as "sleeping in," "working," and "shopping," as in the example shown in Table 6.4. (The table appears later in this chapter.) The presented possibilities are not clearly related to one another and do not bear on a single underlying dimension. For lack of a better term, Schwarz et al. (1992) called sets of this type *nondimensional*. We assume that response order effects in the endorsement of nondimensional response alternatives are primarily a function of their elaboration likelihood if only a few alternatives are presented. If a large number of response alternatives is offered, limitations of respondents' memory may add to the emergence of order effects, as discussed previously.

But other response alternatives bear on a single underlying dimension. For example, respondents are asked, "Which of the following drinks do you consider to be typically German?" and are offered "vodka," "wine," "beer," and so on, as the alternatives. In this case, all response alternatives are to be evaluated along a single dimension. Accordingly, we call them *dimensional*. The dimensionality is not determined by the content of the response alternatives alone but by the respondents' task. If respondents were to report if they ever bought the respective drinks, these drinks would make up a set of nondimensional response alternatives. Evaluating the typicality of these drinks, however, introduces a dimension along which these drinks have to be ordered and gives rise to other judgmental processes that may dilute the impact of cognitive elaboration. We now turn to the nature of these processes, which we already touched upon in Chapters Four and Five.

Asymmetric Contrast Effects

If a given item is preceded by an item that is more extreme on the dimension of judgment, a contrast effect may emerge. For example, respondents are presented a list of persons and are asked to select persons that they like (Ring 1974, 1975). An extremely well-liked person is presented in the middle of a list. In this case, moderately liked persons who are presented in the second part of the list will seem less likable by comparison. Therefore, they will be less frequently selected as "liked" under this order condition. If we compared two orders of this list, the judgmental contrast effect would lead us to conclude that a pronounced primacy effect had emerged. In contrast, if the person presented in the middle of the list were extremely unlikable, the same mechanism of judgmental contrast would increase the endorsement of moderately liked persons presented in the second half of the list. In this case, a comparison of both order conditions would lead us to conclude that a pronounced recency effect had emerged. It should be noted, however, that the underlying cognitive process of judgmental contrast or scale anchoring discussed in Chapter Four is quite different from the cognitive elaboration processes discussed earlier.

A classic example for such a contrast effect was reported by Noelle-Neumann (1970). Respondents were presented a list of food items and were asked to select those that were "typically German." Respondents were more likely to consider a number of food items, including noodles or potatoes, as "typically German" when they were preceded by rice than when they were not. Thus, introducing rice as the first item resulted in pronounced contrast effects in the perception of the other food items. The evaluation of rice itself was unaffected by order manipulations.

While primacy and recency effects in lists are presumably a function of the cognitive elaboration that a given item receives in different positions, contrast effects of the type observed by Noelle-Neumann (1970) are thought to be a function of the items' extremity on the underlying dimension of judgment. Introducing a more

extreme item results in a wider "perspective" regarding the set of stimuli, thus affecting their evaluation as described in Ostrom and Upshaw's (1968) perspective theory, which we reviewed in more detail in Chapter Four. Accordingly, contrast effects should also emerge under conditions where each item is likely to receive about the same degree of attention and elaboration.

To explore this possibility, Schwarz and Münkkel (1988) used a rating rather than a selection task in a laboratory experiment. They asked subjects to rate each of a number of drinks according to how "typically German" they were. As expected, all drinks were rated as more "typically German" if an atypical drink, namely vodka, was presented as the first rather than the last item. In contrast, the rating of vodka was not affected by the order manipulations. Thus, an asymmetric contrast effect emerged, as predicted by Ostrom and Upshaw's (1968) perspective theory. According to that model, respondents use the most extreme stimuli that come to mind to anchor the response scale. In this case, presenting vodka as the first item resulted in a shift of the moderate stimuli away from the anchor. Vodka as the most extreme stimulus in the set, however, is itself unaffected by the order manipulation because the most extreme stimulus is assigned the extreme scores under any order condition except if preceded by a more extreme stimulus.

Moreover, contrast effects of this type do not require that the items be presented on the same list. They have also been shown to emerge if the extreme item is presented as part of a preceding question, provided that this question taps the same dimension of judgment. As an example of this phenomenon, we previously reviewed a study by Schwarz, Münkkel, and Hippler (1990), who asked some respondents to estimate the percentage of Germans who drink vodka and others the percentage who drink beer. Subsequently, all were asked to rate the typicality of various drinks. As expected, respondents who estimated the percentage of Germans who drink vodka rated subsequent drinks as more typically German than subjects who estimated how many Germans drink beer. This replicates the contrast effects obtained when all stimuli were presented on the

same list. But when other subjects were asked as part of the preceding questions to estimate the caloric content rather than the consumption rate of vodka or beer, subsequent typicality ratings were not influenced. Although this question also served to render these drinks highly salient in the interview context, it did not tap the typicality dimension behind the consumption estimates. This finding indicates that contrast effects can emerge as a function of preceding questions if they tap the same underlying dimension of judgment, as we discussed in more detail in Chapter Five.

This emergence of contrast effects bears in important ways on the emergence of primacy and recency effects in general. Specifically, it provides an interesting account for data sets that do not follow our predictions of primacy and recency effects as a function of elaboration and memory processes. If an extremely positive item is presented as part of the stimulus set, it will decrease the endorsement of subsequent moderate items. But if an extremely negative item is presented, it will increase the endorsement of subsequent moderate items. These judgmental effects may lead the researcher to conclude that the data show pronounced recency or primacy effects. Accordingly, the phenomenon of judgmental contrast may dilute the emergence of elaboration phenomena, thus contributing to the mixed findings that characterize this area.

Behavioral Questions

In the preceding discussion, the response alternatives presented as part of an opinion question were portrayed as persuasive arguments. Moreover, it was suggested that the order and mode in which response alternatives are presented influences respondents' opportunity to generate agreeing or disagreeing thoughts. Does this imply that elaboration-based response order effects are limited to opinion questions? We do not believe so. As we shall discuss in Chapter Seven, successful retrieval of information from autobiographical memory (memory about one's behavior) involves several processes that may be affected by the order in which different behaviors are

presented on a list, as Schwarz, Hippler, and Noelle-Neumann (1994) suggested.

First, successful retrieval takes time. Recalling specific events may take up to several seconds (Reiser, Black, & Abelson, 1985) and repeated attempts to recall may result in the retrieval of additional material, even after a considerable number of previous trials (for example, Means, Mingay, Nigam, & Zarrow, 1988; Williams & Hollan, 1981). It is conceivable that respondents' motivation to engage in this effort declines over the course of the list, resulting in more successful retrieval attempts at the beginning than at the end. If so, responses to the first few items should be more accurate than responses to later items, a possibility that would deserve systematic investigation in future research.

Moreover, numerous studies indicate that recall increases as the retrieval cues better match the representation stored in memory (see Strube, 1987, for a review). Most notably, Wagenaar (1986) observed in a study of his own memory that he could retrieve many apparently forgotten events as more specific retrieval cues became available. The retrieval cues presented in standardized lists of daily activities, however, are not very specific. They are most likely to result in successful retrieval when respondents take time to think about them, thus constructing more specific personal retrieval cues from the generic material presented. The likelihood that respondents do this, however, may decrease as a function of motivational deficits. In addition, the respondents' idiosyncratic elaboration of the presented cues may be impaired by the elaborations that they generated in response to preceding cues.

For reasons discussed in the context of opinion questions, the degree of elaboration given to a response alternative is likely to depend on its serial position and its presentation mode (Krosnick & Alwin, 1987). Under visual presentation conditions the first few response alternatives receive more elaboration than the later ones, whereas under auditory presentation conditions the last few receive more elaboration. Therefore, the quality of self-generated retrieval cues may show an interaction effect of serial position and presenta-

tion mode, paralleling the interaction effect discussed for opinion questions. When the list is presented in a visual mode, items presented early in the list should receive more processing, resulting in primacy effects. When the items are read to respondents, however, items presented at the end of the list should receive more processing, resulting in recency effects. To the extent that more elaborate retrieval cues increase the accuracy of recall, we may also expect that retrospective reports are more accurate for early rather than late items in a visual presentation mode. For the same reason, they may be more accurate for late rather than early items in an auditory presentation mode.

To date, none of these hypotheses has been systematically tested. However, they indicate that order- and mode-induced differences in the elaboration of a given response alternative are not limited to opinion question but may affect recall from autobiographical memory as well. Table 6.4 shows an example, taken from Schwarz, Hippler, & Noelle-Neumann, (1994).

In two German surveys conducted by the Allensbach Institute under the direction of Elisabeth Noelle-Neumann (IFD 1008, May 1957; IFD 1022, September 1958), respondents were asked, "Could you please tell me, with the help of this list, what you happened to do last Saturday?" Respondents were provided a list of twenty-eight activities. The order in which the activities were presented was reversed for half of the respondents. Whereas 34 percent of the respondents of the 1957 survey, shown in the first two columns of Table 6.4, reported that they worked at their job when this item was presented first on the list, only 25 percent did so when the item was presented last. Conversely, 15 percent reported that they slept in when this item was presented first, whereas 10 percent reported doing so when the item was presented last. The 1958 data, shown in the second and third columns, replicate this pattern nicely. Primacy effects of up to 9 percentage points emerged when respondents were asked to recall activities they had engaged in last Saturday, that is, a maximum of six days earlier.

These data contradict Carp's (1974) frequently cited conclusion

TABLE 6.4 Reported activities as a function of presentation order.

Activity	Presentation order			
	1957		1958	
	First	Last	First	Last
Working	34%	25%	26%	22%
Sleeping in	15%	10%	17%	12%
N	347	317	318	309

Source: Data based on quota samples of German adults (HFD 1008, May 1957; HFD 1022, September 1958).

Note: Respondents were randomly assigned to one of the two presentation order conditions.

that response order effects are unlikely to emerge for factual reports. In her study, respondents were asked to report the frequency of local trips for various purposes. They were presented with a list of eight frequency alternatives, ranging from "never" to "every day" and the order in which the frequencies were presented was reversed for half of the sample. No effect of response order was obtained. Based on recent research on the impact of frequency scales on retrospective reports (see Schwarz, 1990a, for a review), we assume that frequency scales present a special case. In contrast to categorical response alternatives, where each alternative presents a different substantive option, frequency scales of the type used by Carp present a continuum. As numerous studies have shown (for example, Schwarz, Hippler, Deutsch, & Strack, 1985), respondents use the range of the frequency alternatives as a frame of reference in estimating their own behavioral frequency. Whereas this results in different reports along scales that offer different ranges, the order in which the same range is introduced (for example, "never" to "daily" as opposed to "daily" to "never") has not been found to affect the results (Carp, 1974).

Based on the limited evidence available, we assume that response order effects may emerge on factual reports as well as on opinion questions. We assume that in both cases, they are largely

attributable to differential cognitive elaboration. However, the specifics of the elaboration processes involved are clearly different, reflecting the differences in the nature of attitude judgments and recall tasks.

Verbal Rating Scales

We now turn to the special case of verbal rating scales. In a 1929 paper, Mathews reported the emergence of primacy effects on a verbal rating scale. In a self-administered questionnaire, his subjects were asked to rate different study habits (for example, "take notes during a talk") according to how much they liked them on a scale including "like greatly," "like," "indifferent," "dislike," and "dislike greatly." He observed that "like greatly" was marked 2.9 percent more frequently when it was printed at the extreme left than when printed at the extreme right, whereas "dislike greatly" was marked 2.4 percent more often when it was printed at the extreme left" (Mathews, 1929, p. 130). This basic finding has been replicated in several studies, based on self-administered questionnaires (Belson, 1966; Payne, 1971) and face-to-face interviews (Carp, 1974; Minnigay & Greenwell, 1989; Quinn & Belson, 1969). In all cases, "both negative and positive ends of the scale pulled more responses when presented first, and both pulled fewer responses when presented last," as Carp (1974, p. 584) summarized her findings.

How do we account for these findings? On first glance, it seems that the consistent emergence of primacy effects in face-to-face interviews contradicts the implications of the cognitive elaboration hypothesis, which would predict a recency effect under auditory presentation conditions. However, verbal rating scales do not present different substantive options that respondents need to elaborate on. Rather, terms ranging from "like very much" to "dislike very much" constitute a response continuum where the different options require far less elaboration than is the case with opinion questions of the type shown in Table 6.1. The terms presented on a verbal rating scale can neither be portrayed as persuasive arguments

nor as retrieval cues, in contrast to the material discussed earlier. Similar effects have been obtained on numerical rating scales with labeled endpoints. In general, it seems that ratings are somewhat displaced toward the starting point of the scale. Schwarz and Wyer (1985) suggested that this finding may reflect an anchoring effect of the type described by Tversky and Kahneman (1974). In their study, which we discussed in more detail in Chapter Four, responses were displaced toward the left end of the rating scale when a preceding task induced respondents to proceed from left to right and toward the right end of the rating scale when the preceding task induced respondents to proceed from right to left. Unfortunately, the specific conditions under which anchoring effects of this type are likely to emerge are not yet well understood. At present, we can only note that ratings along a scale are sometimes displaced toward the beginning of the scale, resulting in a primacy effect. Whereas this renders ratings along reversed scales noncomparable, the frequency and size of response order effects along rating scales seems more limited than the frequency and size of order effects that involve discrete response alternatives of the type shown in Table 6.1.

Summary

The considerations reviewed in this chapter suggest that order effects in nondimensional sets of response alternatives are a function of their cognitive elaboration and of the limitations of respondents' memory although memory limitations are likely to be less crucial than has often been assumed. The processes that underlie order effects in dimensional sets of response alternatives, however, are more complex and require a consideration of judgmental contrast in addition to a consideration of cognitive elaboration.

The variations of elaboration likelihood assumptions offered by Krosnick (1992) and Schwarz et al. (1992) provide a rich set of predictions that have only partially been tested. First, the elaboration hypothesis predicts an interaction of serial position and administration mode that is well supported by the available data although

more controlled experiments using the same questions under visual and auditory presentation modes are urgently needed. Second, it predicts conditions under which response order effects should be more or less pronounced, and these predictions have been supported in at least some experiments. Third, it predicts a reversal of response order effects if the presented response alternatives are implausible rather than plausible. Although data bearing on this issue are not yet available, this consideration suggests that response order effects may go in different directions, depending on respondents' attitudes. Accordingly, primacy and recency effects may cancel one another in heterogeneous samples. Moreover, the elaboration likelihood assumption points to a host of other variables—such as respondents' ability and motivation to elaborate on the implications of the response alternatives—that have been explored in research on persuasive communications (see Petty & Cacioppo, 1986a, 1986b; Petty, Ostrom & Brock, 1981, for reviews) but have not been systematically addressed in the context of response order effects (with the exception of cognitive sophistication, see Krosnick, 1992). These predictions deserve more systematic testing, both in sample surveys and in the cognitive laboratory, with a careful assessment of the cognitive responses that are assumed to mediate the observed effects.

In addition, assumptions about the impact of respondents' memory limitations and the emergence of judgmental contrast effects point to conditions under which the impact of cognitive elaboration may be diluted. In respect to memory limitations, our analyses suggest that a straightforward application of findings from the psychological list learning literature cannot account for the pattern of response order effects observed in survey research. Most important, this literature predicts recency effects in the absence of a delay between learning and recall, which is typical for survey research, and provides no mechanism for the emergence of primacy effects under these conditions. Memory limitations, however, may contribute to the size of recency effects under auditory presentation conditions; for example, when the response alternatives are lengthy

and complex. Moreover, they may dilute the size of primacy effects under visual presentation conditions if one assumes that time pressure prevents respondents from going back over the list before reporting their answer, thus introducing some memory burden into the task. Finally, judgmental contrast effects may further dilute the operation of the previously discussed processes if a dimensional set of response alternatives is presented and some response alternatives are considerably more extreme than others.

Accordingly, it comes as little surprise that the emergence of response order effects on opinion questions seems to follow the rule, “sometimes you see them, sometimes you don’t.” When we consider these predictions, it becomes obvious that there is no single main effect. Rather, the emergence and direction of response order effects seems to depend on a complex interaction of serial position, presentation mode; item plausibility, complexity, and extremity; and respondent ability and motivation. Finally, a word of caution. As we noted in the introduction to this chapter, much of the available research is based on flawed designs that confound presentation order with content of adjacent items. Moreover, tightly controlled experiments that provide insight into the assumed underlying cognitive processes are largely missing and our discussion has primarily been based on secondary analyses of archival data. Thus, the conceptualization offered in this chapter is compatible with the general theorizing developed in the context of our discussion of question order effects, which emphasizes the role of cognitive accessibility in the construction of mental representations of the target as a key element of the judgment process. But the empirical support for this conceptualization in the domain of response order is severely limited, requiring more direct tests in tightly controlled experiments.

Practical Implications for Questionnaire Designers

It is clear from the discussion in this and the last two chapters that context effects are ubiquitous and cause complex interactions in response to attitude questions depending on question and response

category order, mode of administration, and information recalled by respondents. As a first step, we believe that researchers need to evaluate the likelihood of context effects when a questionnaire is being designed, so that as far as possible context effects are deliberate rather than unanticipated.

In some cases, context effects are desirable because they help the respondent understand what the researcher wants; they aid in the information retrieval process. In such cases, the researcher should attempt to induce them. This is the equivalent of providing cues for reports of autobiographical behavior.

In other cases, however, the researcher may find the context effect introduces undesirable distortions. The obvious first solution is to eliminate the context by omitting earlier questions or response categories entirely, that is, using open rather than closed questions. If this solution is possible it is clearly the easiest thing to do. But it may be impossible because the data from the earlier questions are required or because the response categories are necessary for the respondent to be able to understand the question.

In personal interviews, the introduction given by the interviewer to a question or series of questions can be used to increase, reduce, or eliminate context effects, as desired by the researcher. In self-administered questionnaires, the same results can be obtained by printing instructions, by putting the questions together in a box on one page, or by putting questions on separate pages.

Other ways of reducing context effects involve the question itself. The more general or ambiguous the question, the more likely context effects will occur. It usually makes sense to keep the question as specific and unambiguous as possible. As we have pointed out, context effects will be much smaller or may vanish completely if the respondent already has a judgment or substantial amounts of relevant information chronically accessible in memory. This is not something the researcher can manipulate, but it needs to be kept in mind, for example, when the data from news polls that ask for respondent reactions to newly developing political events are interpreted.

Within a question, if response alternatives are necessary, their

order may be randomized unless there is some natural ordering. However, randomization “eliminates” order effects only in a rather mechanical sense. By averaging across the effects generated by different orders, randomization simply ensures that our conclusions do not reflect one specific order. Nevertheless, any given respondent’s answer will be affected by the order in which the response alternatives or questions were presented and the numerous different orders generated by randomization will increase the variance in the sample, potentially affecting statistical analyses. Technically, the use of computer-assisted interviewing makes it far easier to randomize the order of alternatives. An added advantage of computer interviews, even if self-administered, is that the respondent is aware only of the questions that precede a given question and not those that follow. Similarly, within a section of related questions in a questionnaire, it may be possible to randomize the order of the questions using computer-assisted interviewing, again assuming there is no natural ordering.

Finally, it is best to avoid items or persons that achieve extremely positive or negative ratings when asking for judgments about a series of items because those that follow will be strongly affected. Thus, if Adolf Hitler or Mother Teresa are included among a list of political personages, the ratings of a current president or prime minister will be strongly influenced based on their position in the list.

7

Autobiographical Memory

Survey researchers frequently ask respondents questions about their past behavior. In so doing, researchers are asking respondents to search a particular part of their memory that has come to be called *autobiographical memory*. Brewer (1986, 1994) distinguishes three types of memories that can properly be called autobiographical—*personal memories* (for example, having mental images of doing a particular thing, such as visiting the Taj Mahal), *autobiographical facts* (remembering that one visited an elderly aunt in St. Louis when one was ten years old but having no mental image of the event), and *generic personal memory* (having a mental image of what it is like in general to drive from Boston to New York but having a generic view rather than an image of any specific point in the drive). Generic personal memory is distinguished from semantic memory, which is knowledge of facts that are not experienced in time or place (for example, that Columbus discovered America in 1492) and from generic perceptual memory (for example, having a visual image of the state of California).

Autobiographical memory has become the subject of much recent work in cognitive psychology. (For a textbook introduction see Conway, 1990; for reviews see Bradburn, Rips, & Shevell, 1987; Neisser & Winograd, 1988; Rubin, 1986; Schwarz, 1990a.) In this chapter, we consider some of the approaches to the study of